n = 19 tau6 native per-shift solve, protocol v2 (frozen, not run)

2026-10-19

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Status (2026-10-19): frozen; no timed run has been launched. This protocol replaces v1. Its Tier 1 run, 39307409409, was invalid because its S0 self-test failed. v1's protocol and archive are unchanged. The workflow `n19-native-tau6-v2.yml` has `workflow_dispatch` as its only trigger, so committing never launches a run. The owner decides when to dispatch it.

Why v1 failed, and what v2 changes

The full diagnosis is in `diagnosis/README.md`.

- **Cause.** v1's `arm.sage` binding read `run.sage`'s `MODULUS`, whose coefficients run lowest degree first, as highest degree first. It passed the reciprocal polynomial (829707) instead of Sage's modulus (854355). The native solve and its brute force share the context, so they agreed with each other on 300 of 300 systems and with Sage on 293.
- **Not affected.** `tau.sage`, the baseline row and `lib.rs`'s solve are correct.

	v1	v2
binding	inside `arm.sage`, which only Sage can load; modulus reversed	`native_binding.py`, plain Python, also run by the local checks; modulus lowest degree first
field check	none	in every native process, after its intervals: `tau6_modulus` = Sage's `F.modulus()`, and 256 `tau6_mul` products = Sage's
S0	300 systems; Sage vs native, native vs its own brute force; counts only	the 7 v1 mismatch systems + 300 new ones (Sage, native, brute force and direct S3 in Sage's field) + all 2,375 multi-solution systems (Sage order = native order); every system recorded
variety order	ascending (u, w), matches Sage on 176 of 2,375 multi-solution systems	ascending (w, u), Sage's order (2,375 of 2,375); ordered decompositions gated B = N = Naa = NX
`cargo test`	3 tests	7: adds the 7 systems, the v1 failure reproduced over the reciprocal, 15 recorded Sage orders, and the product through the C ABI
panel	`n19-native-heldout-v1`	`n19-native-heldout-v2`: new seed; also excludes the v1 panel
arm order, bootstrap	`Random(2026101919 + i)`, `default_rng(20261019)`	`Random('...-v2-20261019:order:%d')`, `default_rng(20261020)`
trigger	push (path filter) and dispatch	dispatch only

Everything else is v1's design, unchanged. This covers the question, the arms B, N, Naa, NX and rho, the intervals, the primary R = mean cold CPU(B) / mean cold CPU(N) with its paired bootstrap, the decision rule, the A/A window [0.95, 1.05], the planted lower bound > 1, O1 and the per-point gates. See `protocol.json` and v1's README for the derivation.

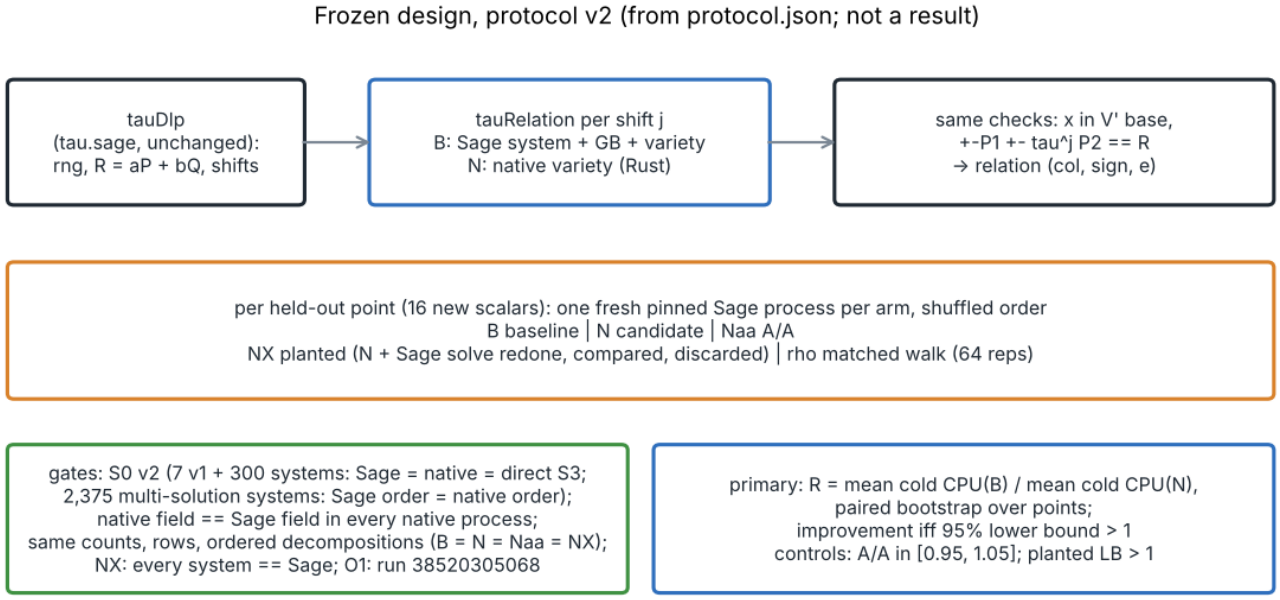


Figure 1: Frozen v2 design (not a result). Source: make_figures.py; vector: figures/design.svg.

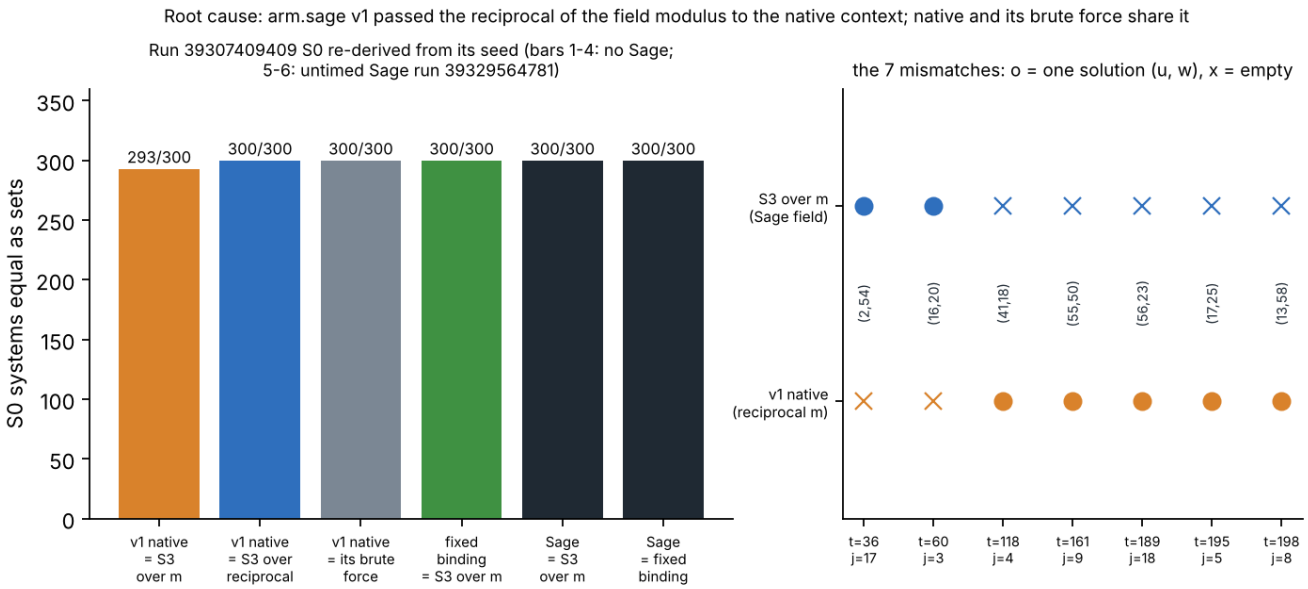


Figure 2: Run 39307409409's S0, re-derived without Sage. Vector: figures/s0_diagnosis.svg.

Order of a multi-solution variety (new in v2)

Sage's `variety()` returns a multi-solution variety ascending in (w, u). The untimed Sage diagnostic run 39329564781 recorded this on all 2,375 such systems of the field. v1 returned ascending (u, w), which matches Sage on only 176 of them.

- **Why it now matters.** v1's panel met only shift-0 swaps, where the order is harmless. The v2 panel meets three non-swap systems, and Sage's order differs from (u, w) on two of them. B and N would have returned different relations there.
- **The change.** v2's native solve writes the variety in Sage's order. The enumeration is unchanged.
- **The new gates.** The ordered decompositions must be identical across B, N, Naa and NX. S0 re-checks the order on all 2,375 systems on the runner. NX must agree in order on every multi-solution call.

See `diagnosis/README.md`.

Local checks before freezing (cloud-agent VM, no Sage; not timing evidence)

- **`cargo test --release --locked`.** 7 of 7 pass, locally and on the diagnostic runner.
- **Sage, untimed** (run 39329564781, `diagnosis/sage_run_39329564781/`). On the 307 v2 S0 systems, Sage = native = direct S3 on 307 of 307 (4 non-empty). `diagnosis/order_check.json` checks the order: the v2 library's output equals Sage's recorded order on 2,375 of 2,375 multi-solution systems and on all 607 S0 systems.
- **S0 without Sage.** `diagnosis/s0_v2_local.py` runs the v2 binding on the 307 v2 S0 systems. Native, its brute force and the pure-Python direct S3 agree on 307 of 307 (4 non-empty, 0 multi). The field check passes, and with the v1 modulus it fails on 256 of 256 products (`diagnosis/s0_v2_local.json`).
- **O1 and the panel without Sage.** `smoke/local_v2.py` runs `tau.sage`'s control flow in pure Python with the v2 binding. On the ten archived scalars it reproduces run 38520305068's tau6 targets, `gb_calls`, `spurious`, relations and logs on 10 of 10. On the 16 v2 points all logs verify, with a mean of 5,653 solve calls. Across both parts the candidate meets 36 multi-solution calls. 4 of them, on points p09, p10 and p11, hit 3 distinct non-swap systems (`smoke/local_v2.json`).
- **Power** (`analysis/power.json`, same model as v1, seed 20261020). With 16 points, power is 1.00 at a true ratio of 1.20, and the rejection rate with no effect is 0.035.

Files

- **Code.** `native/` (solve and tests), `native_binding.py`, `arm.sage`, `run_schedule.py`, `verify_receipt.py` and `stats.py`.
- **Frozen inputs.** `make_protocol.py` → `protocol.json`; `make_panel.py` → `inputs/panel.json`; `make_schedule.py` → `schedule.json`; `power_check.py` → `analysis/power.json`.
- **Diagnosis.** `diagnosis/`. The untimed Sage diagnostic is `diagnosis/sage_diag.sage`, run by `.github/workflows/n19-native-s0-diagnostic.yml`.
- **Timed workflow.** `.github/workflows/n19-native-tau6-v2.yml` (dispatch only).
- **Report.** `make_figures.py`, `figures/`, `README.pdf` and `report_header.tex`.